MINI PROJECT REPORT

LIBRARY MANAGEMENT SYSTEM USING JAVA AND MYSQL

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Course: BCA (Hons.)

Subject: DBMS

Date of Submission: 27 October 2026

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1. INTRODUCTION

The Library Management System is a simple application developed using Java and MySQL. It allows users to view the available books, search for books by title, and add new books to the database.

The application uses JDBC (Java Database Connectivity) to establish a connection between the Java application and the MySQL database. Book information entered through the application is stored directly in the database.

2. OBJECTIVE

The objectives of this mini project are:

- To develop a basic Library Management System using Java.
- To create and manage a database using MySQL.
- To establish connectivity between Java and MySQL using JDBC.
- To insert, retrieve, and search for book records.
- To verify that data entered through the application is stored in MySQL.

3. TECHNOLOGIES USED

Technology	Purpose
Java	Application development
Apache NetBeans	Java development environment
MySQL	Database management
JDBC	Connecting Java to MySQL

MySQL Connector/J JDBC driver for MySQL

4. DATABASE DESIGN

Database Name: library_db

The database contains the following tables:

1. Books: Stores book ID, title, author, publisher, publication year, ISBN, and available copies.

2. Members: Stores member ID, name, email, phone number, and membership date.

3. Transactions: Stores borrowing and returning details and links books with members.

The Books table is used by the current Java application to add, view, and search for books.

Screenshot 1: MySQL Database and Tables

```
1  CREATE DATABASE library_db;
2  USE library_db;
3
4  CREATE TABLE Books (
5      book_id INT PRIMARY KEY AUTO_INCREMENT,
6      title VARCHAR(150) NOT NULL,
7      author VARCHAR(100) NOT NULL,
8      publisher VARCHAR(100),
9      year_published INT,
10     isbn VARCHAR(20) UNIQUE,
11     available_copies INT NOT NULL DEFAULT 0
12 );
13
```

5. APPLICATION–DATABASE CONNECTIVITY

The application connects to the MySQL database using JDBC. The connection is established through the DBconnection.java class.

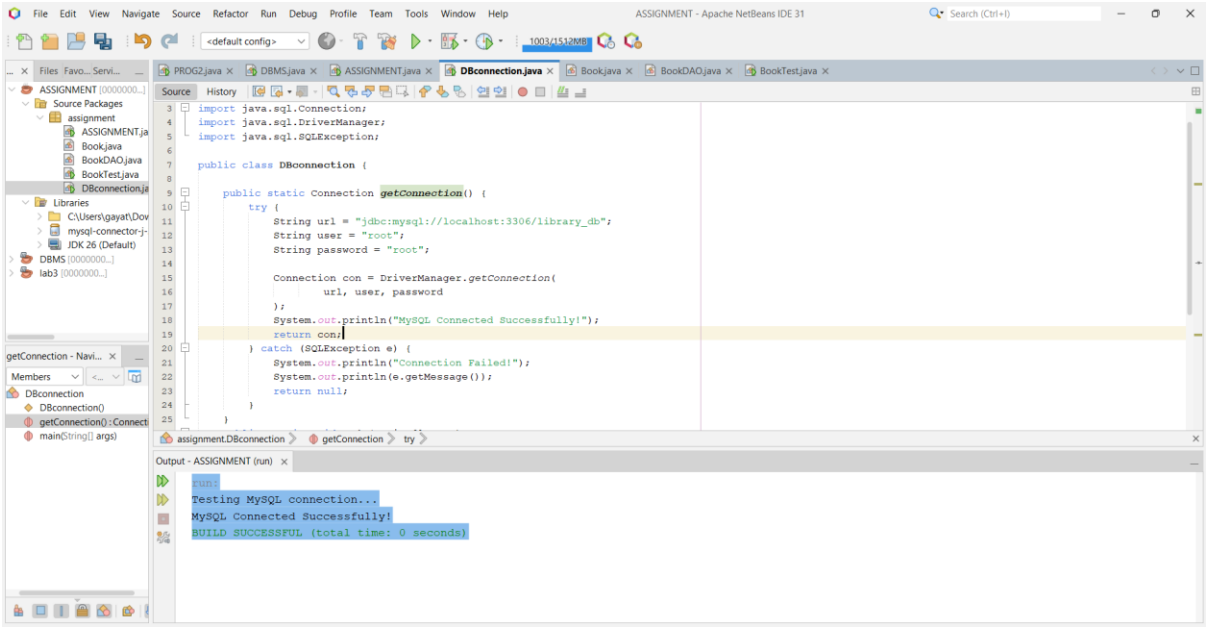
The database URL used is:

jdbc:mysql://localhost:3306/library_db

When the connection is successful, the application displays the message:

MySQL Connected Successfully!

Screenshot 2: Successful Database Connection



6. APPLICATION INTERFACE AND FUNCTIONALITIES

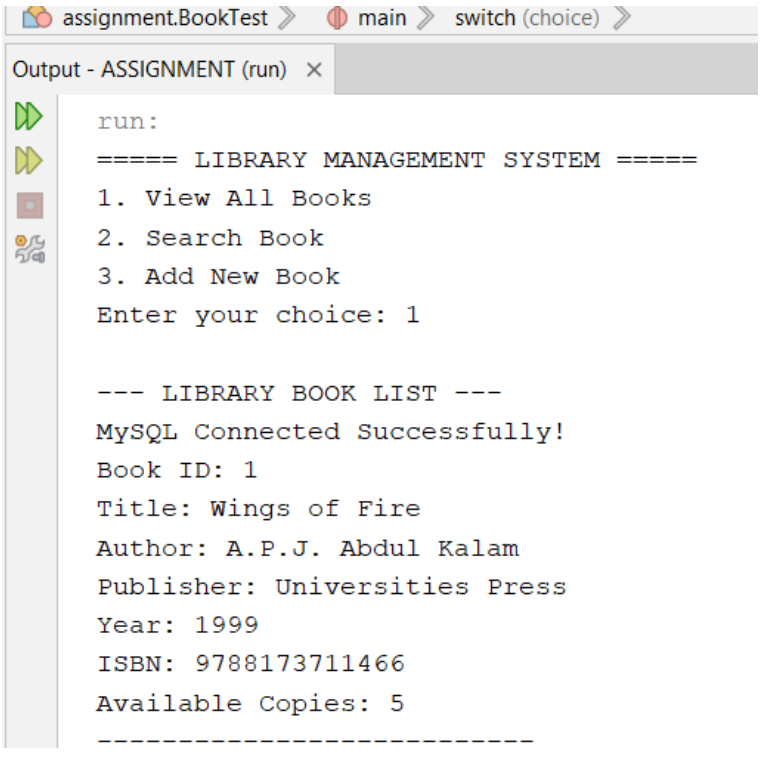
The application provides a console-based menu with three options:

1. View All Books
2. Search Book
3. Add New Book

6.1 View All Books

This function retrieves all book records stored in the Books table using the SQL SELECT statement. The application displays the book ID, title, author, publisher, publication year, ISBN, and available copies.

Screenshot 3: View All Books

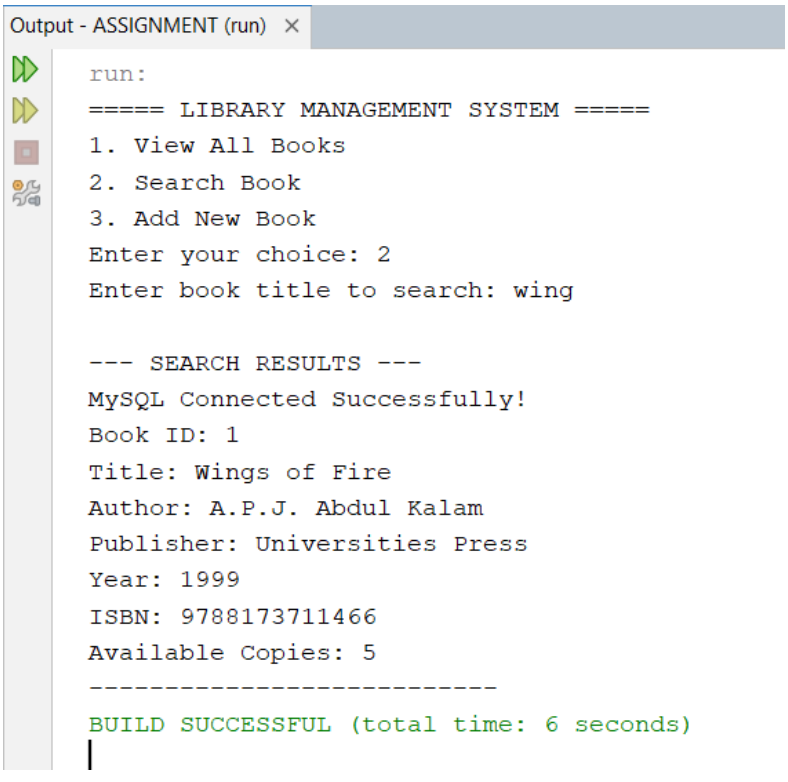


```
Author: Paulo Coelho
Publisher: HarperOne
Year: 1988
ISBN: 9780061122415
Available Copies: 3
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Book ID: 3
Title: Atomic Habits
Author: James Clear
Publisher: Avery
Year: 2018
ISBN: 9780735211292
Available Copies: 4
---------------------------
Book ID: 4
Title: Famous Five
Author: Enid Blyton
Publisher: Penguin Books
Year: 1980
ISBN: 978-0-13-149508-1
Available Copies: 10
---------------------------
BUILD SUCCESSFUL (total time: 5 seconds)
```

6.2 Search Book

This function allows users to search for a book by entering its title or part of its title. The application retrieves matching records from MySQL using a parameterized SQL query.

Screenshot 4: Search Book

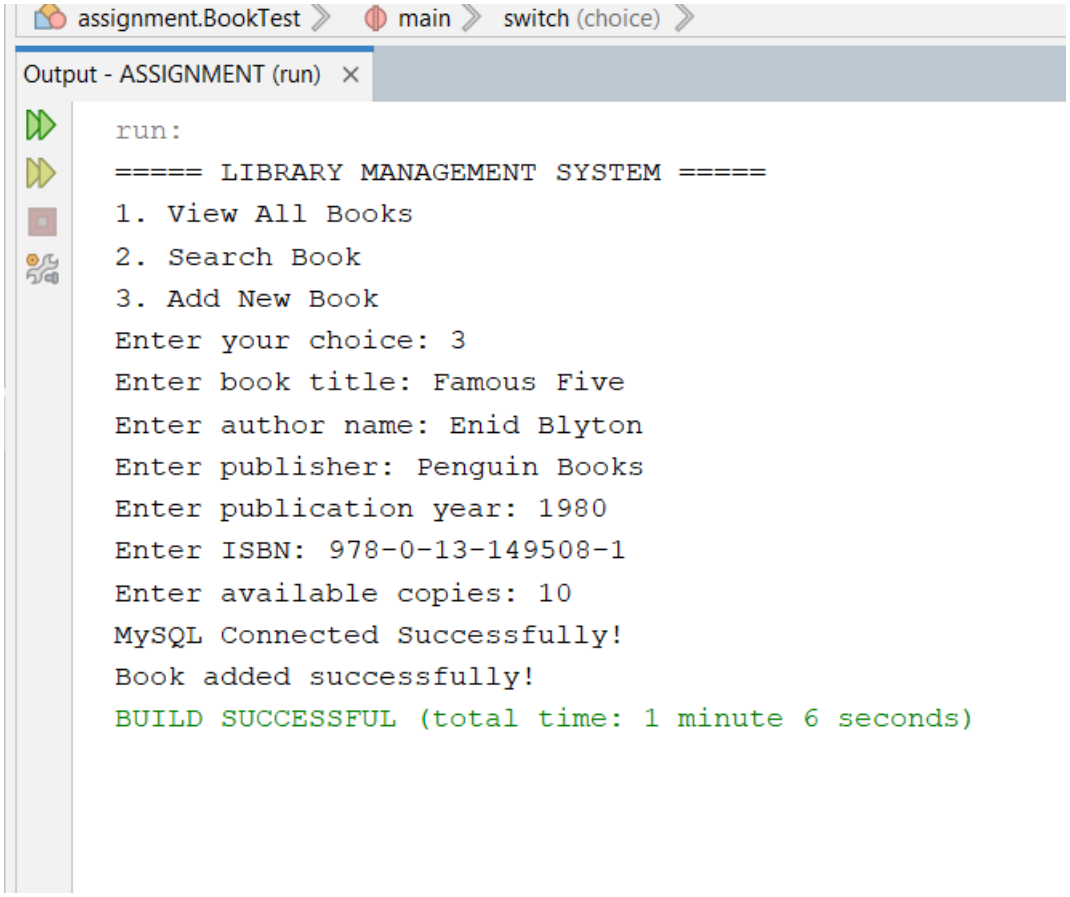


6.3 Add New Book

This function allows users to enter a book's title, author, publisher, publication year, ISBN, and available copies. The details are inserted into the Books table using an SQL INSERT statement.

The application displays a success message when the record is inserted successfully.

Screenshot 5: Add New Book



7. VERIFICATION OF DATA IN MYSQL

To verify database connectivity, a new book was added through the Java application. The Books table was then queried in MySQL Workbench to confirm that the newly inserted record was stored in the database.

SQL query used:

SELECT * FROM Books;

The record entered through the application appeared in the MySQL query results, confirming that the Java application can insert data into the database.

Screenshot 6: Book Record Verified in MySQL

[INSERT SCREENSHOT HERE — MySQL Workbench showing the book added through the Java application]

```
58
59 •  USE library_db;
60 •  SELECT * FROM Books;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	author	publisher	year_published	isbn	available_copies
1	Wings of Fire	A.P.J. Abdul Kalam	Universities Press	1999	9788173711466	5
2	The Alchemist	Paulo Coelho	HarperOne	1988	9780061122415	3
3	Atomic Habits	James Clear	Avery	2018	9780735211292	4
4	Famous Five	Enid Blyton	Penguin Books	1980	978-0-13-149508-1	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL

8. CONCLUSION

The Library Management System was developed using Java, JDBC, and MySQL. The application successfully connects to the MySQL database and supports adding new books, viewing all books, and searching for books by title.

The insertion of book records through the Java application was verified in MySQL Workbench. This project demonstrates the basic integration of a Java application with a relational database using JDBC.

9. PROJECT FILES

The project contains the following Java source files:

- DBconnection.java — establishes the database connection.
- Book.java — represents book information.
- BookDAO.java — performs database operations.
- BookTest.java — provides the application menu and user interaction.

The SQL script contains the database setup, table creation statements, sample data, and verification queries.